Critical path

When slack is bad, or even when it is good, you still need the path

Goldens to remember

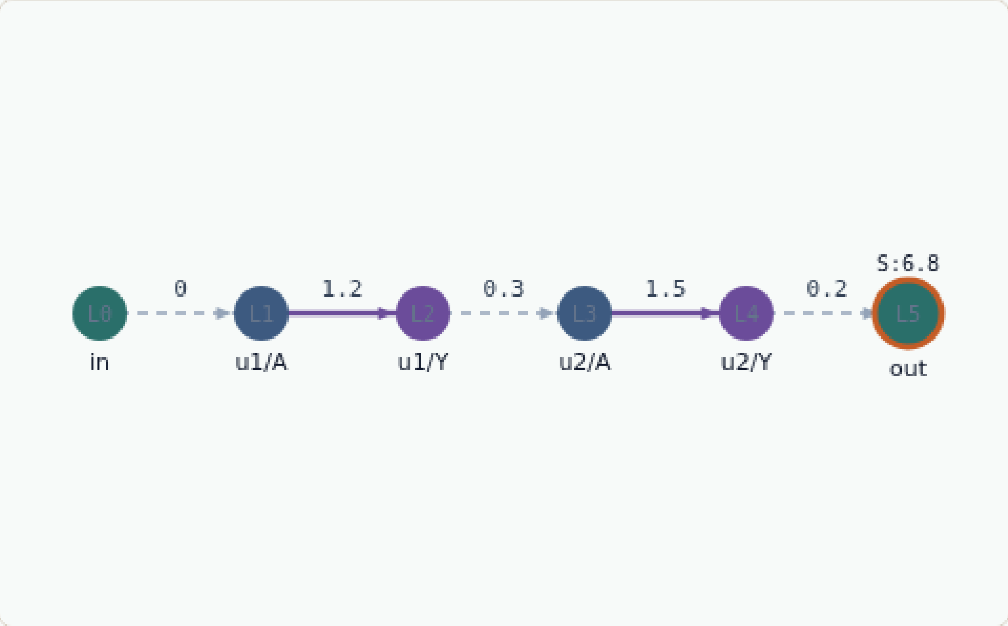
- Critical path has six pins
- Path delay equals A(out)=3.2
- Always match A(u)+d to A(v)
- Keep these numbers handy, the browser challenges and Track A tests use the same instance
- <!-- algorithm-walkthrough -->

Start from the worst endpoint

CRITICAL PATH

Start from the worst endpoint

Step 1 / 5 · worst-sink · module02-03-critical-path



Explanation

Critical-path traceback begins at the endpoint with the worst setup slack. Here that is out with slack 6.8—still the only sink.

- Pick the worst slack sink
- Trace into arrival tags
- Do not guess from the source

```
sink: out
setup slack: 6.8
```

Start from the worst endpoint

Step to the matching predecessor

CRITICAL PATH

Step to the matching predecessor

Step 2 / 5 · match-pred · module02-03-critical-path

Explanation

At each pin, choose a predecessor u where A(u) + delay equals A(v). That arc is on the critical path.

- Match A(u)+d to A(v)
- Ties: prefer larger A(u)
- Walk until a source

```
out ← u2/Y (+0.2)
u2/Y ← u2/A (+1.5)
```

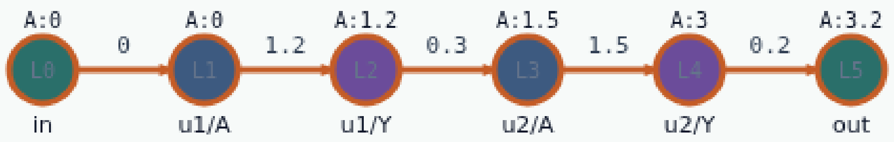
Step to the matching predecessor

The full golden path

CRITICAL PATH

The full golden path

Step 3 / 5 · full-path · module02-03-critical-path



Explanation

The complete critical path is in → u1/A → u1/Y → u2/A → u2/Y → out. Arrival at out equals the sum of arc delays along this path: 3.2.

- Six pins on one path
- Path delay = A(out)
- Golden list matches PROP_GOLDENS

```
in → u1/A → u1/Y → u2/A → u2/Y → out
path delay = 3.2
```

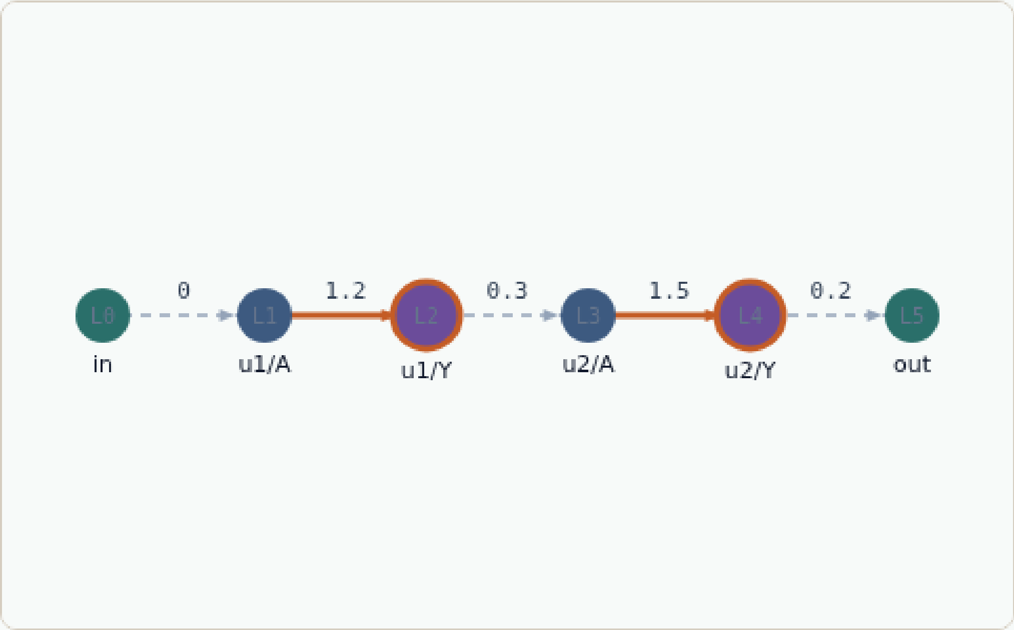
The full golden path

Fix the path, not a random gate

CRITICAL PATH

Fix the path, not a random gate

Step 4 / 5 · why-it-matters · module02-03-critical-path



Explanation

Optimization and ECO work chase critical paths. Tag matching keeps you honest when reconvergence exists—on this chain there is only one route.

- Trace before you resize
- Reconvergence needs care
- Toy chain has one route

```
reconvergence: none
cells on path: u1, u2
```

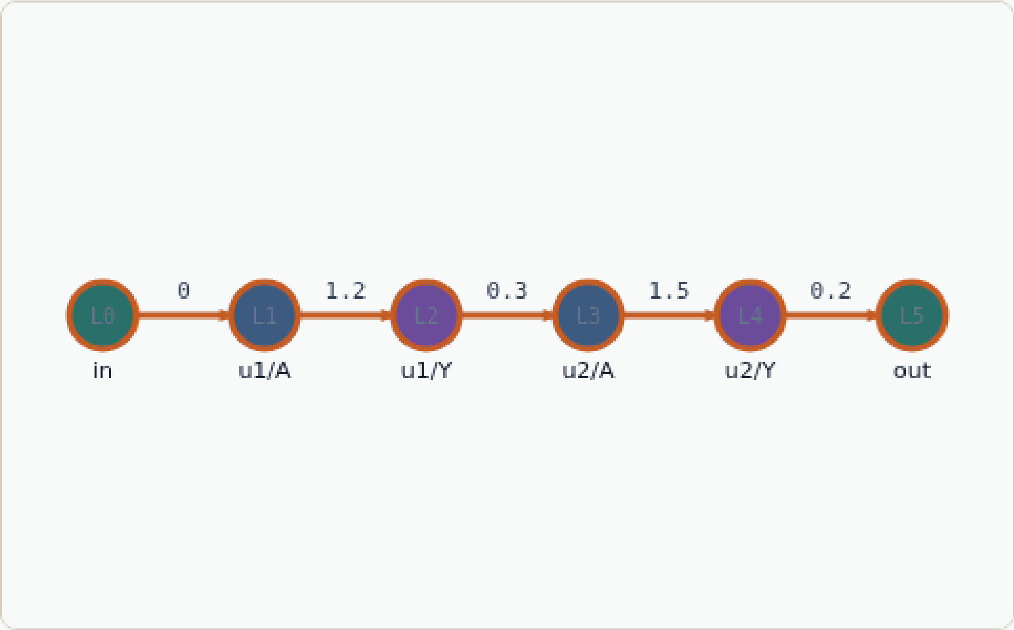
Fix the path, not a random gate

Slack and path travel together

CRITICAL PATH

Slack and path travel together

Step 5 / 5 · slack-path · module02-03-critical-path



Explanation

Reports pair worst slack with its path. Next labs edit delays and exceptions—always re-trace after the tags change.

- Slack names the problem
- Path names where to look
- Re-trace after edits

```
slack 6.8 @ out
path length 6
```

Slack and path travel together

Browser lab track

- In the browser lab, open **critical-path**
- Load the starter, run the analysis once, and read the metrics panel
- Orient yourself, challenge panel, canvas, Check, then mirror the same goldens in code

Implement track

- In the implement track
- Run `python3 common/test_propagate.py` (and the timing-graph test) until the goldens print

Pitfall

- Do not mix setup and hold required maps
- Do not propagate before the graph is levelized
- After an edit or exception, recompute, stale tags lie

Your turn

- Finish the checklist on at least one track, preferably both
- When your numbers match the goldens, take the quiz, then continue